Wigner Resolution from Action Capacity

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Smoothing the Wigner function gives a non-negative phase-space portrait of a quantum state, with the smoothing scale conventionally left free. We show that the scale is fixed by the state. The state's covariance defines a Heisenberg cell of action capacity $\pi\,\Delta x\,\Delta p$, admitting a family of de Gosson quantum blobs deforming at constant action $h/2$. Convolving the Wigner function with the kernel of a single $h/2$ blob renders it non-negative. Integrating over the family resolves the portrait at $\pi\hbar^2/(\Delta x\,\Delta p)$, the symplectic polar dual of the Heisenberg cell. The portrait is everywhere non-negative and carries the structure of the Wigner function undisplaced. Wigner resolution carries no free parameter, set by the state's action capacity alone and sharpening as that capacity grows.

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The Wigner function represents a quantum state in phase space [1, 2]. It is defined pointwise below Planck's quantum of action, where Hudson's theorem forces it negative [3]. Its negativity is a resource for quantum advantage [4, 5] and a measure of nonclassicality [6]. A non-negative distribution supports classical sampling [7–11] and follows from smoothing the Wigner function at a freely chosen scale. Here the state's action capacity $\pi\,\Delta x\,\Delta p$ sets that scale without a free parameter.

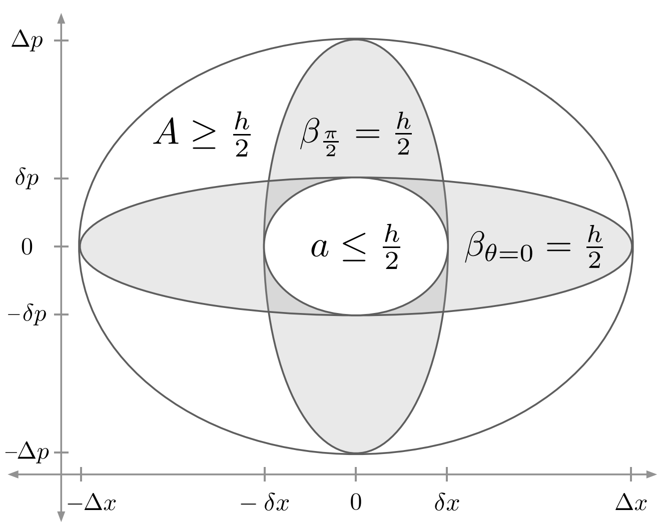

FIG. 1. **Heisenberg cell, quantum blobs at principal axes, and resolution cell.** The Heisenberg cell is the state's covariance ellipsoid with action capacity $A = \pi\,\Delta x\,\Delta p \geq h/2$. The quantum blob family (Fig. 2) has two members at the principal axes, $\beta_{\theta=0} = \pi\,\Delta x\,\delta p = h/2$ and $\beta_{\pi/2} = \pi\,\delta x\,\Delta p = h/2$. Zurek's relations $\delta p \sim \hbar/\Delta x$ and $\delta x \sim \hbar/\Delta p$ locate the semi-axis scales from the state's action capacity. The resolution cell a is an ellipse with semi-axes $\delta x, \delta p$ and action $a = \pi\,\delta x\,\delta p = \pi\hbar^2/(\Delta x\,\Delta p)$. A circumscribes $\beta_{\theta=0}$ and $\beta_{\pi/2}$, which circumscribe a.

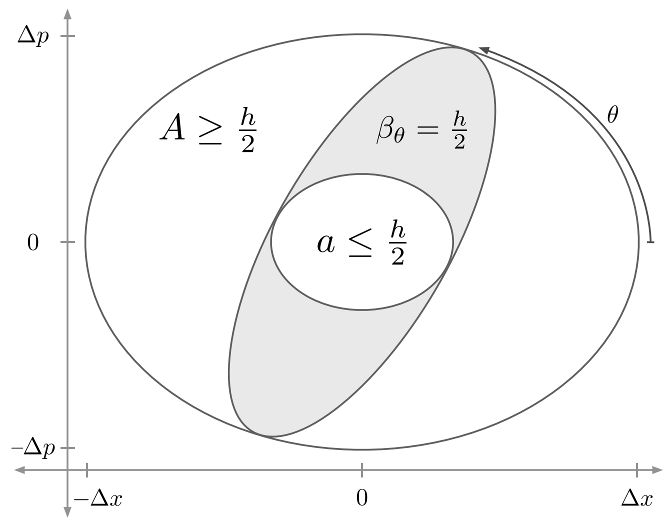

FIG. 2. **Heisenberg cell, quadrature family of quantum blobs, and resolution cell.** The Heisenberg cell A circumscribes the quantum blob family $\{\beta_\theta\}$, which circumscribes the resolution cell a. Across θ the family deforms at constant action $\beta_\theta = h/2$. At the principal angles $\theta = 0$ and $\theta = \pi/2$, the family recovers $\beta_{\theta=0}$ and $\beta_{\pi/2}$ from Fig. 1.

Planck divided phase space into elementary regions of action [12]. Heisenberg's uncertainty principle $\sigma_x\,\sigma_p \geq \hbar/2$ [13–15] fixes their minimum area. Geometrically, de Gosson gave this action region the form of the state's covariance ellipse, with symplectic capacity $\pi\,\Delta x\,\Delta p$ [16] and semi-axes $\Delta x = \sqrt{2}\,\sigma_x$ and $\Delta p = \sqrt{2}\,\sigma_p$. This is the Heisenberg cell (Figs. 1, 2) of action capacity,

$$A = \pi\,\Delta x\,\Delta p \geq h/2. \tag{1}$$

The Wigner function carries structure at small scales. Zurek located those scales, $\delta p \sim \hbar/\Delta x$ and $\delta x \sim \hbar/\Delta p$, from the state's action capacity [17]. We identify them as the principal axes of a family of de Gosson quantum blobs (Fig. 1),

$$\beta_{\theta=0} = \pi\,\Delta x\,\delta p = h/2, \qquad \beta_{\pi/2} = \pi\,\delta x\,\Delta p = h/2. \tag{2}$$

The capacities $\Delta x, \Delta p$ run along the major axes, the resolutions $\delta x, \delta p$ along the minor.

Between these principal members the family $\{\beta_\theta\}$ deforms across $\theta \in [0, \pi)$ at constant action $h/2$ (Fig. 2). The Heisenberg cell A circumscribes each β_θ. In the

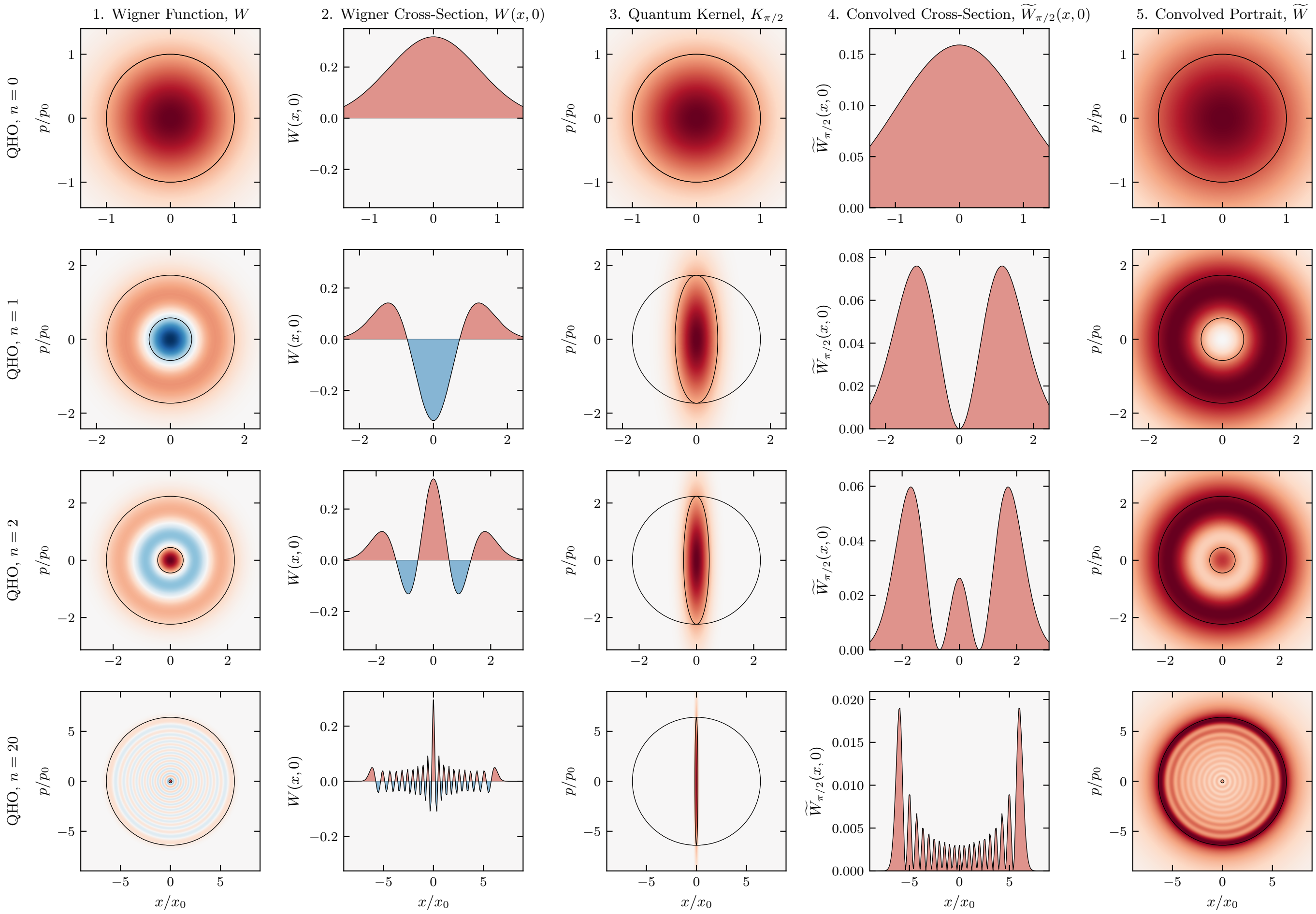

FIG. 3. **Harmonic oscillator.** Rows correspond to Fock states $n = 0, 1, 2, 20$ with action capacity $A/(h/2) = 2n+1$. Column 1: Wigner function $W(x,p)$ (negativity in blue) overlaid with the Heisenberg cell A and resolution cell a. Centered at the covariance origin, a marks the resolution of the finest structure (the Zurek scale) of the bare, unconvolved Wigner function. Column 2: Cross-section $W(x,0)$ exhibiting negative dips. Column 3: Quantum kernel $K_{\pi/2}$ overlaid with A and the quantum blob $\beta_{\pi/2}$ (action $h/2$, restoring Planck's quantum of action). Column 4: Convolved cross-section $\widetilde{W}_{\pi/2}(x,0)$ at quadrature angle $\theta = \pi/2$, which is non-negative. Column 5: The portrait $\widetilde{W}(x,p)$, integrated over the blob family across all θ, overlaid with A and a (symplectic polar duals). It is everywhere non-negative, with positive lobes at the positions of those in W, resolved at a. For the vacuum ($n = 0$), A and a coincide as a Gaussian blob at $A = h/2$. As n increases, A grows and a reciprocally shrinks ($a\,A = (h/2)^2$). At $n = 20$, the rings condense toward a classical energy trajectory, leaving a as a fine central cell. Axes are in the oscillator scales $x_0 = \sqrt{\hbar/m\omega}$ and $p_0 = \sqrt{\hbar m\omega}$.

scaled coordinates $X = x/\Delta x$, $P = p/\Delta p$ it is one ellipse turned through θ,

$$\beta_\theta = \left\{(x,p) : X_\theta^2 + \tfrac{A^2}{(\pi\hbar)^2}\, P_\theta^2 \le 1\right\}, \tag{3}$$

inscribed in the unit disk. The quadratures $X_\theta = X\cos\theta + P\sin\theta$ and $P_\theta = -X\sin\theta + P\cos\theta$ are those of homodyne tomography at phase θ.

The Wigner function of the state is

$$W(x,p) = \frac{1}{\pi\hbar}\int \psi^*(x+s)\,\psi(x-s)\,e^{2ips/\hbar}\,ds. \tag{4}$$

The Wigner function of each quantum blob β_θ is the centered Gaussian whose $h/2$ ellipse is β_θ,

$$K_\theta(x,p) = \frac{1}{\pi\hbar}\exp\!\left[-X_\theta^2 - \tfrac{A^2}{(\pi\hbar)^2}\, P_\theta^2\right]. \tag{5}$$

The convolution of W with each quantum kernel K_θ is

$$\widetilde{W}_\theta(x,p) = (W * K_\theta)(x,p). \tag{6}$$

Convolving W with the kernel K_θ of a single quantum blob β_θ lifts it to Planck's quantum of action and yields a non-negative function [8, 9, 18, 19].

Zurek also located the scale $a \sim \hbar \times (\hbar/A)$ [17, 20] of the chessboard structure in the compass state. We identify its rotationally invariant form as the resolution

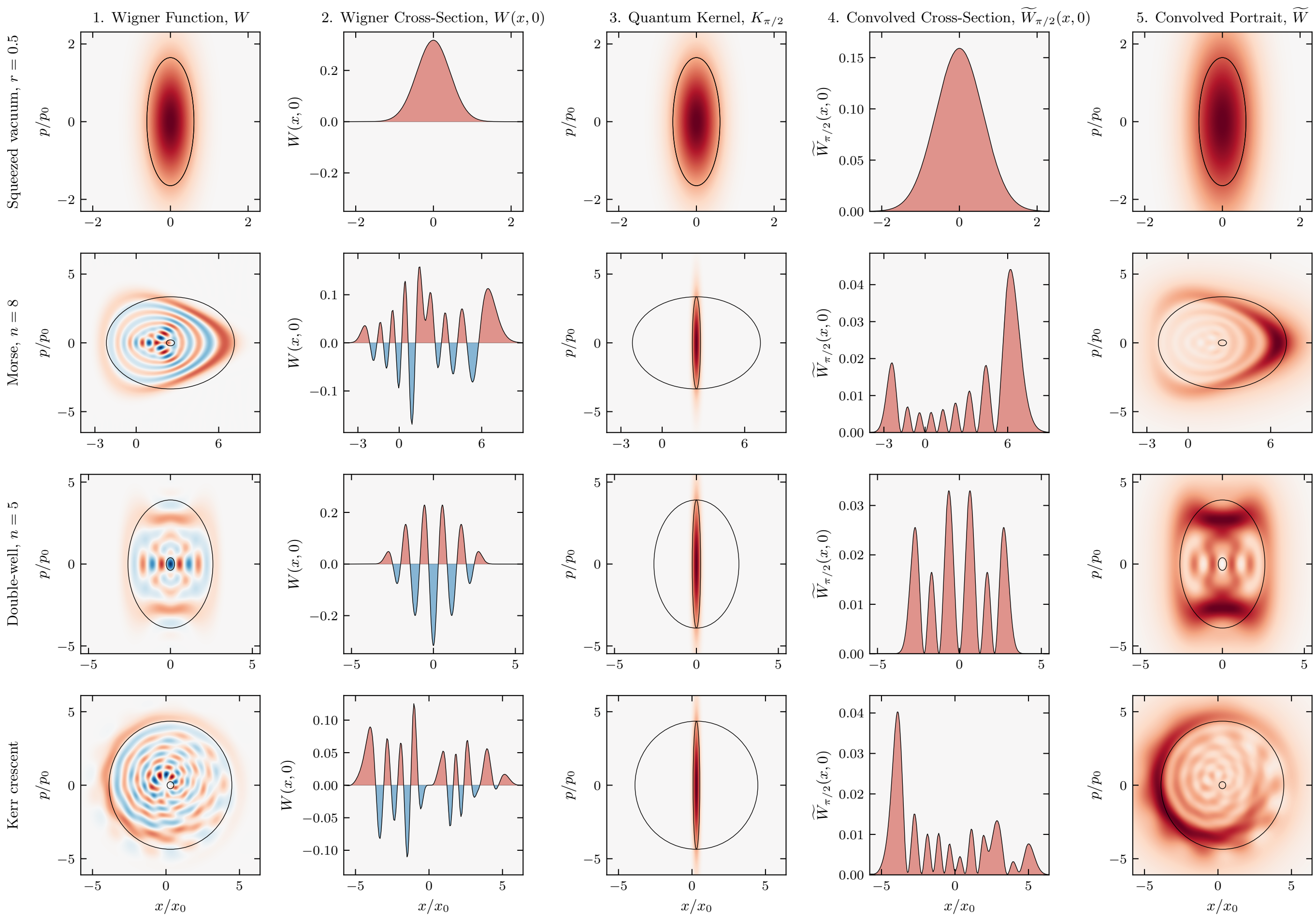

FIG. 4. **Squeezed vacuum, eigenstates, and the Kerr crescent.** Rows correspond to the squeezed vacuum ($r = 0.5$), the Morse eigenstate ($n = 8$), the double-well eigenstate ($n = 5$), and the Kerr crescent, a coherent state sheared at constant photon number. Columns are as in Fig. 3. The Kerr state is shown in the principal frame of its covariance, where the cell overlays are axis-aligned. The portrait $\widetilde{W}$ (Column 5) remains strictly non-negative across all four states, resolved at a.

cell a (Fig. 5, column 1). It is the ellipse with semi-axes $\delta x, \delta p$ (Figs. 1, 2) and action

$$a = \pi\,\delta x\,\delta p = \frac{\pi\hbar^2}{\Delta x\,\Delta p}. \tag{7}$$

It is the resolution of the state's finest structure in the bare Wigner function (Figs. 3, 4, 5, column 1) and the symplectic polar dual of the Heisenberg cell A [21, 22]. The duality is reflexive, so A and a each fix the other, and antimonotone, so a shrinks as A grows. Each blob β_θ is circumscribed by A and circumscribes a, nesting as $A \supseteq \beta_\theta \supseteq a$ (Fig. 2). Resolution and capacity are reciprocal at the quantum of action, $a\,A = (h/2)^2$.

A single blob commits to a quadrature the state does not favor. Integrating $\widetilde{W}_\theta$ over the family favors none, yielding the phase-space portrait,

$$\widetilde{W}(x,p) = \frac{1}{\pi}\int_0^\pi \widetilde{W}_\theta(x,p)\,d\theta. \tag{8}$$

Each $\widetilde{W}_\theta \geq 0$, so $\widetilde{W} \geq 0$ everywhere. Where standard tomography inverts the marginals to recover the Wigner quasi-probability W [23–25], convolving W with $\{K_\theta\}$ over the same θ builds the non-negative portrait $\widetilde{W}$.

The convolution and the integral are linear, so the family reduces to a single convolution,

$$\widetilde{W} = W * K, \qquad K = \frac{1}{\pi}\int_0^\pi K_\theta\,d\theta, \tag{9}$$

and the breadth of the kernel's Fourier transform $\widehat{K}$ sets the portrait's resolution. With (ξ_x, ξ_p) the chord conjugate to (x,p), integrating over θ bounds $\widehat{K}$ above by a single Gaussian,

$$\widehat{K}(\xi_x, \xi_p) \leq \exp\left[-\tfrac{1}{4}\left(\delta x^2\,\xi_x^2 + \delta p^2\,\xi_p^2\right)\right], \tag{10}$$

with equality at the origin. The bound cuts off at δx and δp, so the portrait carries no structure finer than $\pi\,\delta x\,\delta p = a$, derived exactly in the Supplemental Material [26].

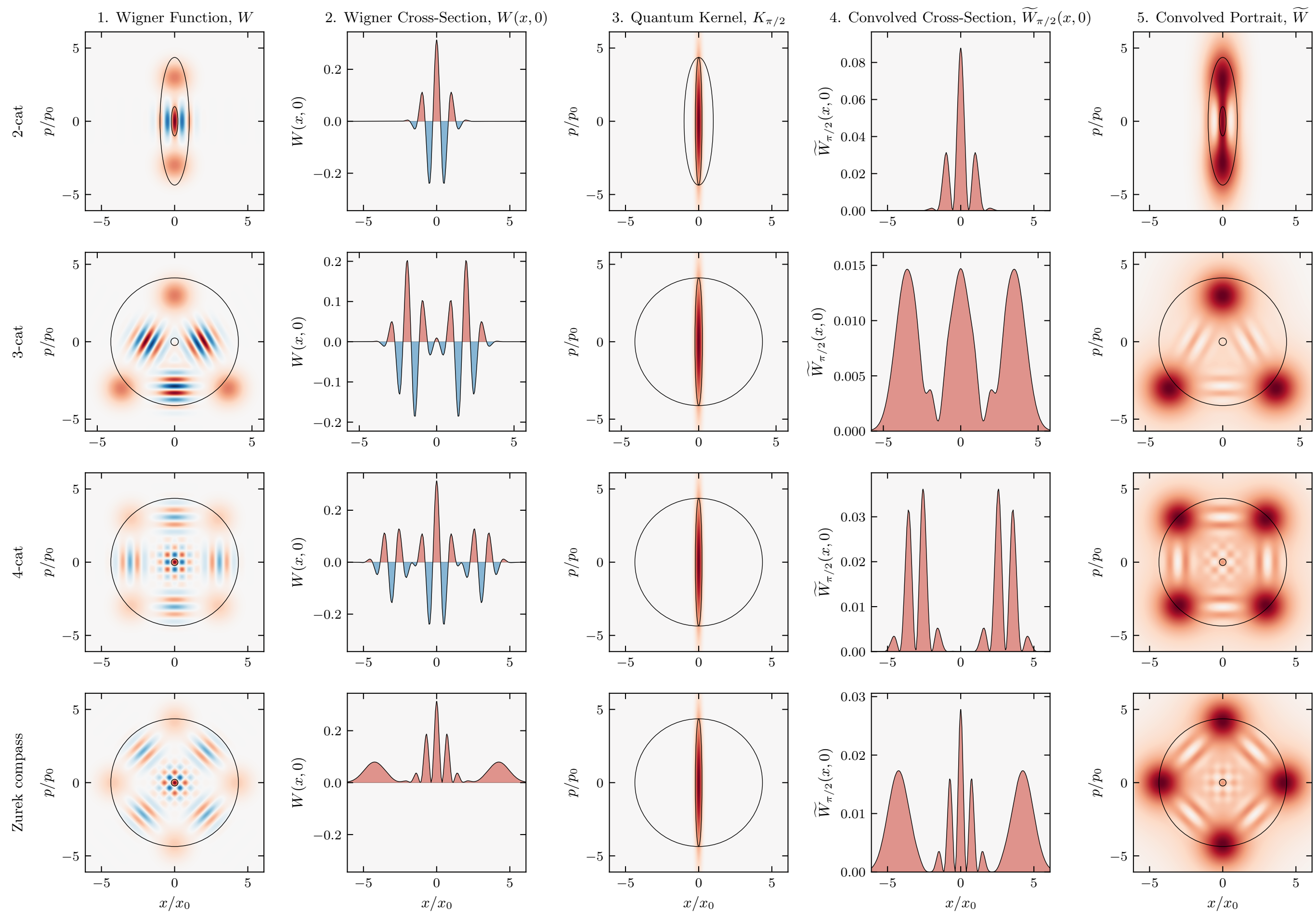

FIG. 5. **Cat states.** Rows correspond to the 2-cat, 3-cat, 4-cat, and Zurek compass states. Columns are as in Fig. 3. As before, the portrait $\widetilde{W}$ (Column 5) is everywhere non-negative, with positive lobes at the positions of those in W, resolved at a. The 4-cat and Zurek compass states differ by a $\pi/4$ rotation in phase space.

A quantum blob β_θ too large for A to circumscribe, or too small to circumscribe a, is a blob of another state and portrays that state, not this one [21, 27]. Every $h/2$ blob of this state circumscribes the resolution cell a, so no $h/2$ smoothing of this state resolves finer, and a kernel below $h/2$ fails to render the portrait non-negative [8, 9]. The finest resolution is a, reached by the family $\{\beta_\theta\}$ across $\theta \in [0, \pi)$.

The principal frame, where the Heisenberg cell A is axis-aligned, costs no generality. The Wigner function transforms covariantly under the symplectic group [19], so the map that diagonalizes a state's covariance carries it to this frame. The Kerr crescent, a sheared coherent state, is resolved this way (Fig. 4). In one degree of freedom, A fixes the family uniquely, and the Zurek structure is already present for states with $A > h/2$. In higher dimensions, action $h/2$ no longer singles out a blob [27], symplectic capacity and volume diverge, and the higher-dimensional case remains for future work.

A fixed vacuum-width Gaussian [7] smooths every state equally, matching only the vacuum $A = h/2$ and erasing the detail of larger states $A > h/2$. Methods with a free parameter [28–30] resolve more finely, the scale set from outside the state. Here the state sets its own scale. Its action capacity A fixes the resolution cell a as the symplectic polar dual, with nothing left free.

Figures 3, 4, and 5 show numerical results across twelve states. The portrait $\widetilde{W}$ is non-negative to machine precision, the chord-domain transfer carrying W to $\widetilde{W}$ follows Eq. (10), and the cross-spectral phase between W and $\widetilde{W}$ vanishes. Across every state the portrait $\widetilde{W}$ resolves W at a and displaces nothing [26].

The uncertainty product $\sigma_x \sigma_p$ grows unbounded, the micro-to-macro transition Heisenberg [31] and Bohr [32] identified as requiring care. The quantum blob does not. As A grows, the family $\{\beta_\theta\}$ deforms at fixed action $h/2$. Squeezing deforms a state by external apparatus [33]. Here the scale called sub-Planck [17] is the minor axis δ of an $h/2$ blob, narrowing as the conjugate capacity Δ grows. A Schrödinger cat with capacity $\Delta p \sim 1$ kg·m/s has resolution $\delta x = \hbar/\Delta p \sim 10^{-34}$ m, nineteen orders sub-nuclear [34]. The correspondence principle follows from $A \to \infty$ and $a \to 0$, the physical limit underlying the heuristic $h \to 0$. The resolution sharpens as the action capacity grows, $a\,A = (h/2)^2$. We read this structure as the geometry of phase space itself, which de Gosson finds underlying the uncertainty principle [16, 18, 21].

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